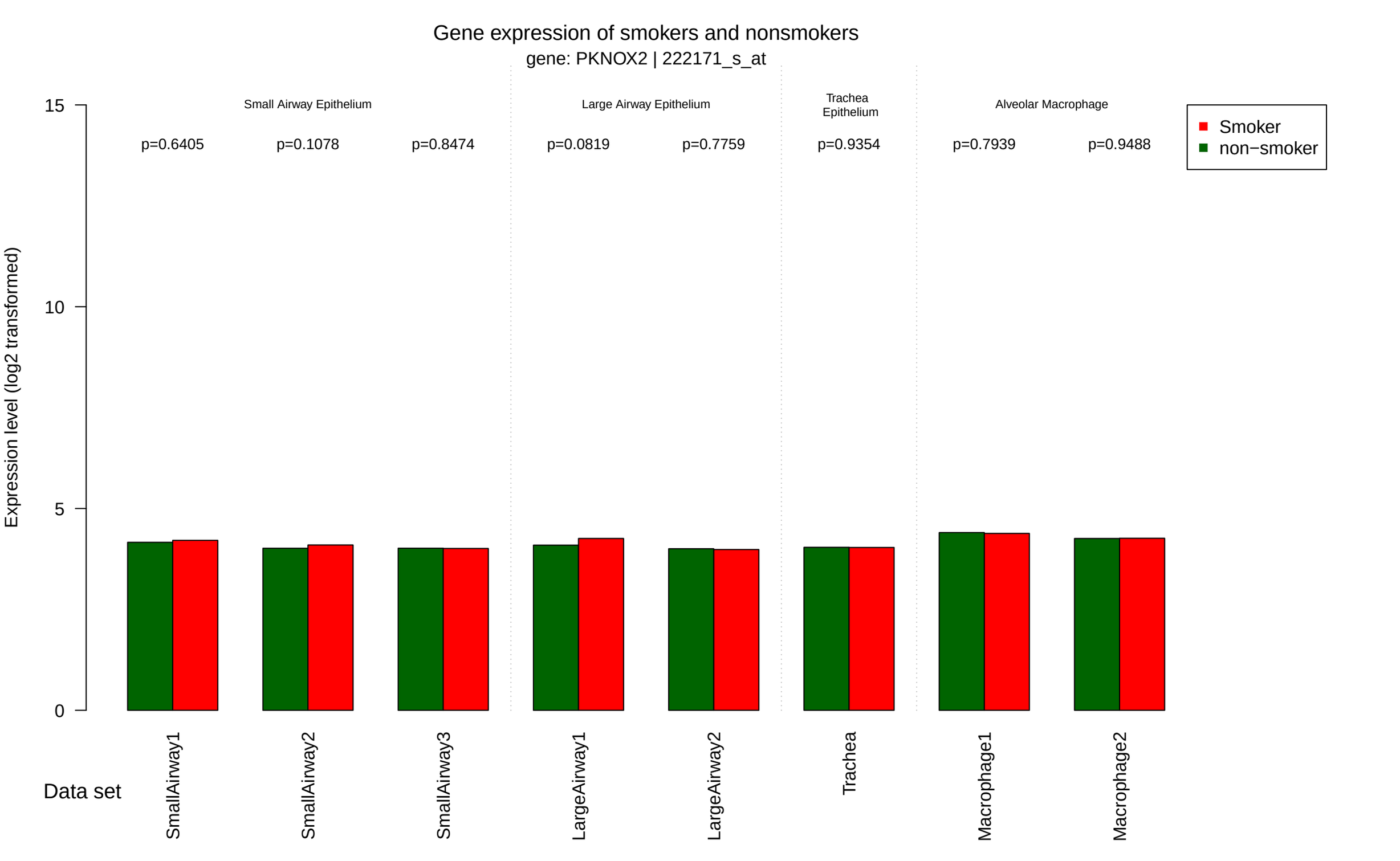
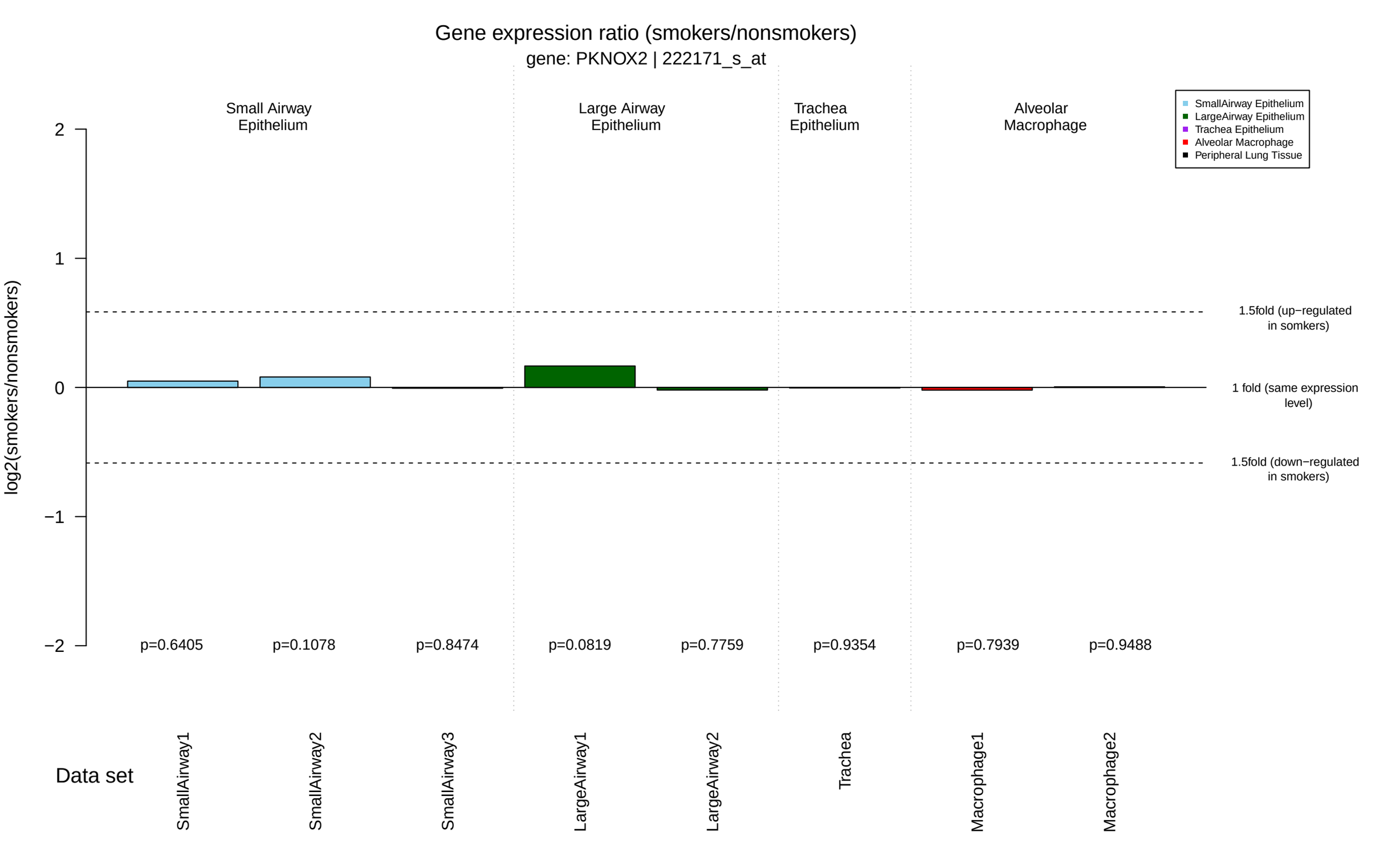
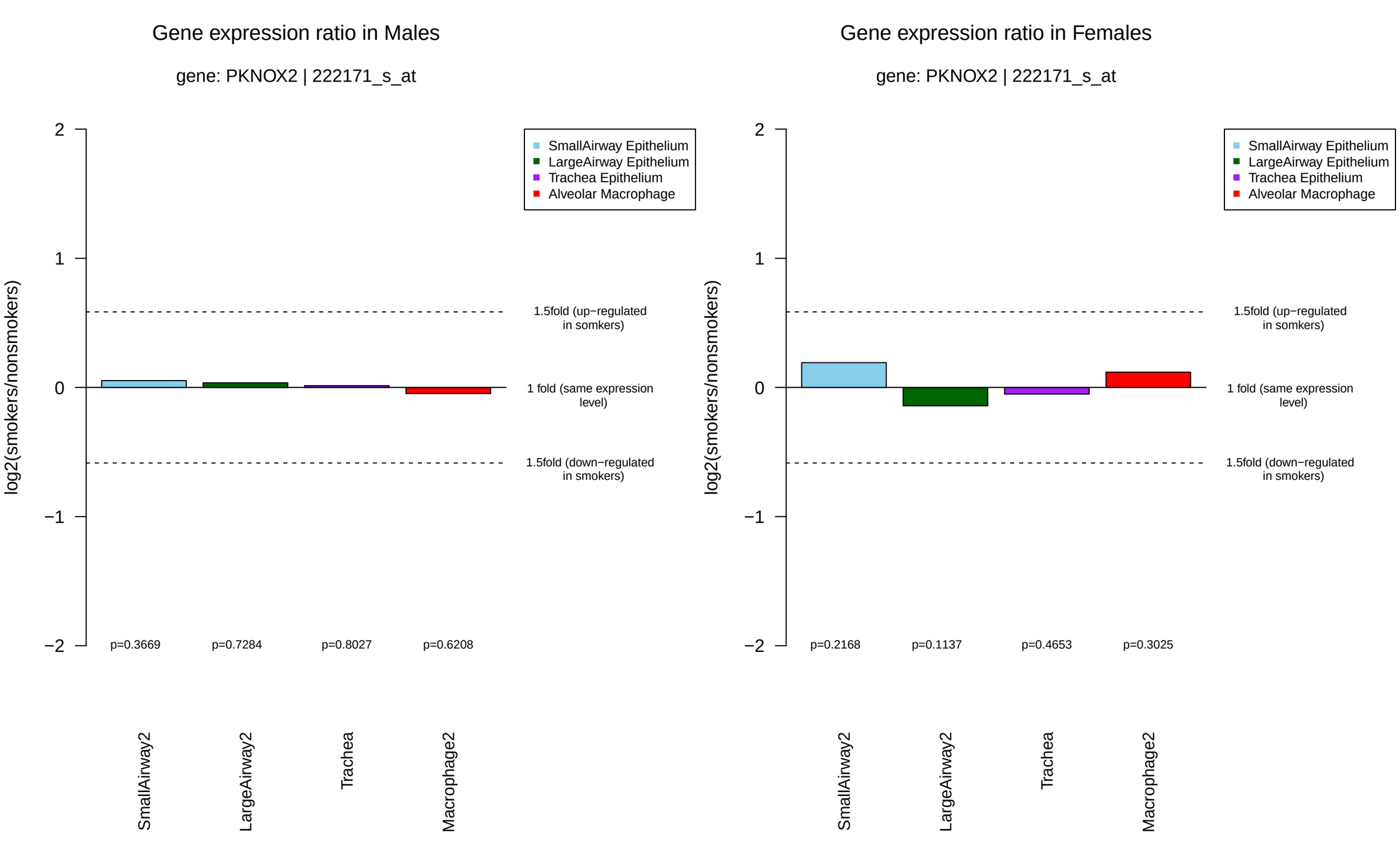
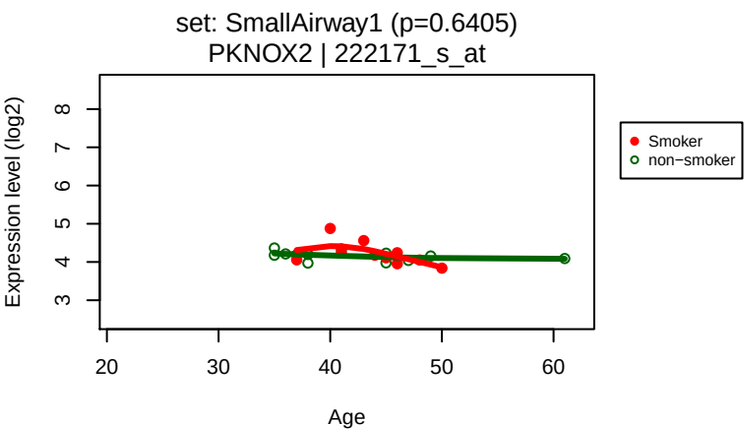
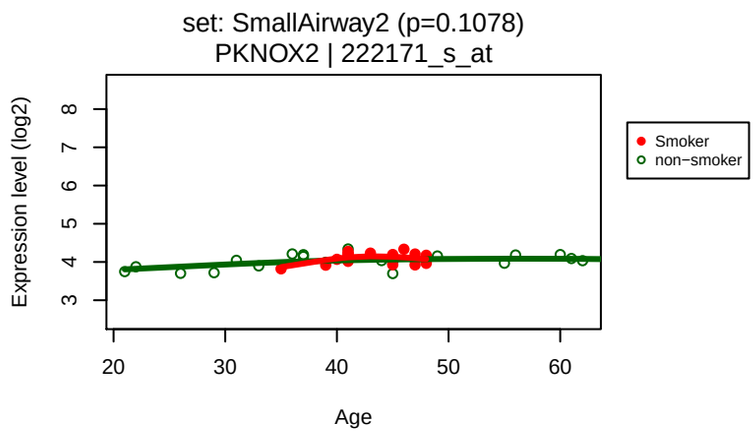
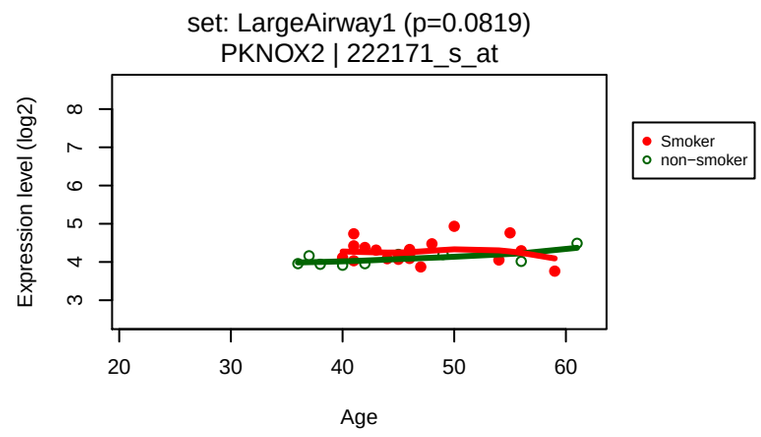

Gene expression of smokers and nonsmokers
gene: PKNOX2 | 222171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6405
p=0.1078
p=0.8474
p=0.0819
p=0.7759
p=0.9354
p=0.7939
p=0.9488
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

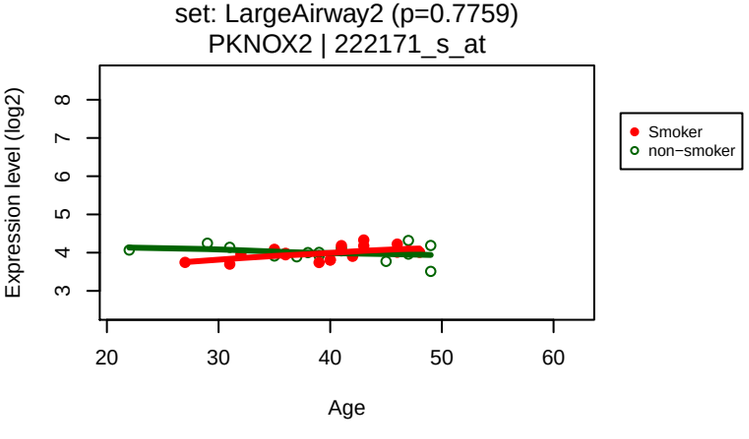

Gene expression ratio (smokers/nonsmokers)
gene: PKNOX2 | 222171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6405
p=0.1078
p=0.8474
p=0.0819
p=0.7759
p=0.9354
p=0.7939
p=0.9488
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

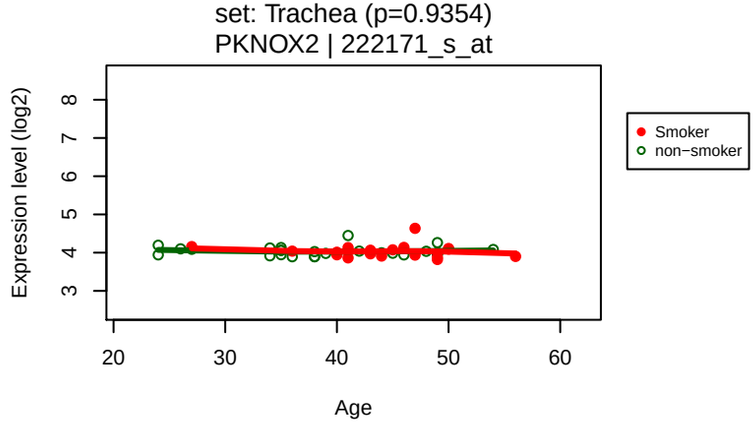

Gene expression ratio in Males
gene: PKNOX2 | 222171_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3669
p=0.7284
p=0.8027
p=0.6208
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PKNOX2 | 222171_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2168
p=0.1137
p=0.4653
p=0.3025
SmallAirway2
LargeAirway2
Trachea
Macrophage2

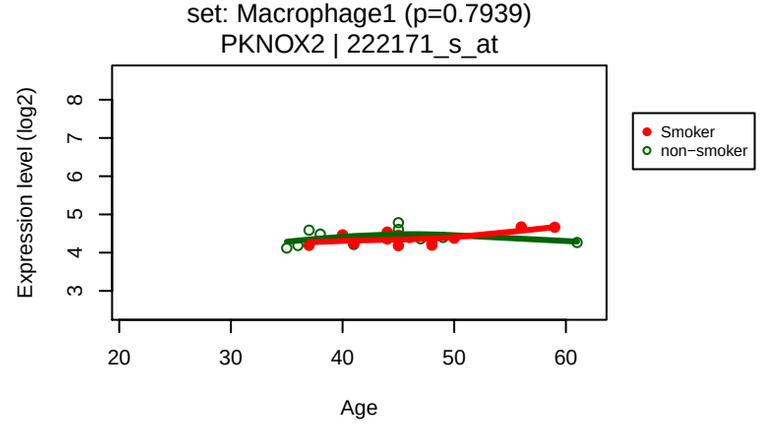

set: SmallAirway1 (p=0.6405)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker

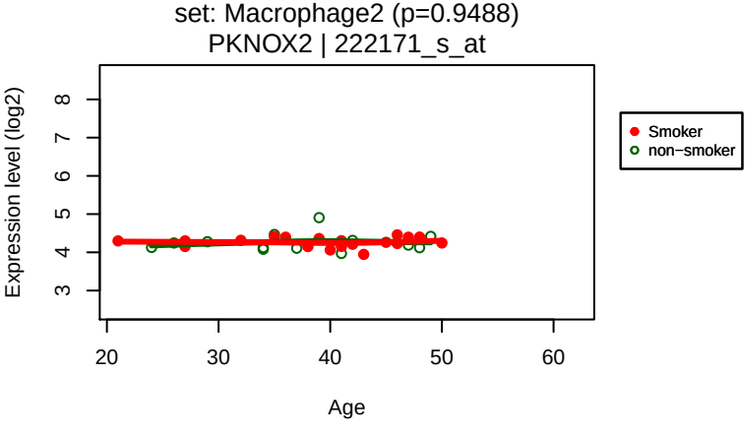

set: SmallAirway2 (p=0.1078)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker

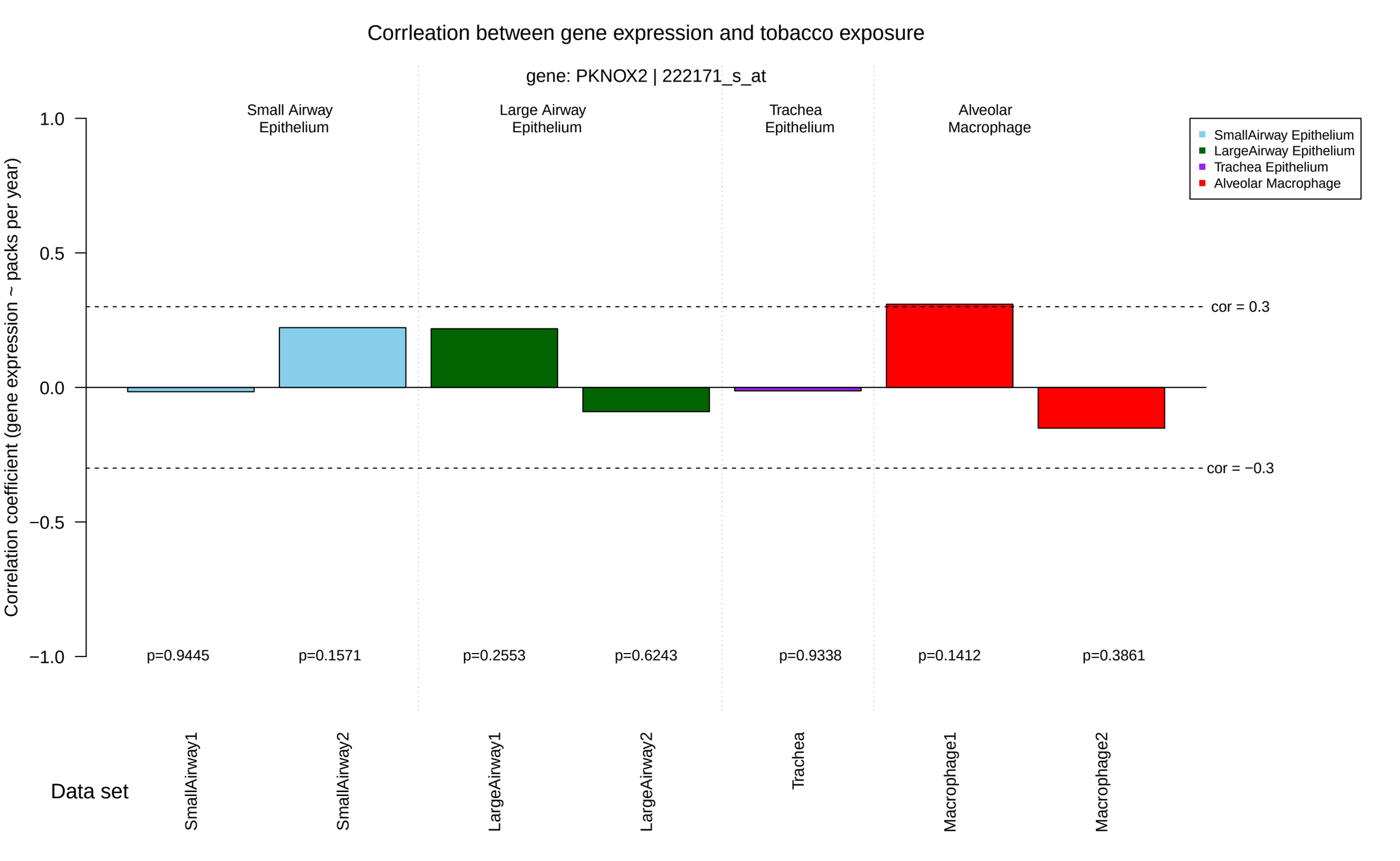

set: LargeAirway1 (p=0.0819)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7759)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9354)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7939)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9488)
PKNOX2 | 222171_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PKNOX2 | 222171_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9445
p=0.1571
p=0.2553
p=0.6243
p=0.9338
p=0.1412
p=0.3861
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2